

Activity: Computer Forensics

Part I. File Carving

1. Look at the data on the file system (Click on Data Sources and look at the hex values on the right). The file system has no files, but why are we able to find items on the disk image? Explain why the file system has no files but there are items that can be found on the disk image.

Ans:

When a user issues a delete command, the operating system only deletes or destroys the metadata and reports that the space is now “free”. However, deleting metadata does not mean that the actual content of the file (file content) is removed from the disk sectors. The raw data of the file still exists in the “free” space until new files are written over it.

2. How many objects can you find?

Ans: 14

3. List all the objects here and report on whether or not the content is accessible or damaged/corrupted. Also note which files were actually already deleted.

Ans: Only image and text files can be opened. Audio files cannot be opened in Autopsy, but they are accessible in external tools

The screenshot shows the Autopsy 4.22.1 interface. The left sidebar displays the 'Data Sources' tree with 'File System (0)' selected. The main pane shows a table of 14 results. The table columns are: Name, S, C, O, Modified Time, Change Time, Access Time, Created Time, Size, Flags(Dir), and Flags(Meta). The files listed are:

Name	S	C	O	Modified Time	Change Time	Access Time	Created Time	Size	Flags(Dir)	Flags(Meta)
✗ #0000281_Nick_is_a_pretty_man_with_a_2003_docu...	0	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	19968	Unallocated	Unallocated
✗ #0000321.wmv	0	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	8037267	Unallocated	Unallocated
✗ #0016021.wav	0	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	318894	Unallocated	Unallocated
✗ #0016693.xls	0	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	23040	Unallocated	Unallocated
✗ #0016741_Prudent_Engineering_Practice_for_Crypto	0	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	1399508	Unallocated	Unallocated
✗ #0019477.pdf	0	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	122434	Unallocated	Unallocated
✗ #0019717.jpg	0	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	29885	Unallocated	Unallocated
✗ #0019777.jpg	0	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	444314	Unallocated	Unallocated
✗ #0020645.jpg	0	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	99298	Unallocated	Unallocated
✗ #0020841.gif	0	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	5498	Unallocated	Unallocated
✗ #0020853_mcoov.mov	0	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	550653	Unallocated	Unallocated
✗ #0021929.wmv	0	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	1036994	Unallocated	Unallocated
✗ #0023957.ppt	0	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	11264	Unallocated	Unallocated
✗ #0023981_wword60.zip	0	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	78899	Unallocated	Unallocated

4. Think securely: If we want to delete files on a magnetic hard disk and not have them be recovered by any tool, what do we need to do? And how much time do you think you need to wipe a 1TB magnetic hard disk?

Ans: Writing over the disk at least 7 times will take approximately 14 hours, or using a strong magnet for a few seconds.

5. Will file carving be able to recover deleted files on an SSD? Why or why not?

Ans: File carving cannot recover deleted files on an SSD, or it will be very difficult. SSDs use the TRIM or UNMAP command, and the SSD controller responds by clearing the actual data.

Part II. Investigation

1. List all directories that were traversed in 'RM#2'.

Ans: CarvedFiles

The screenshot displays the Autopsy 4.22.1 interface. The left sidebar shows the 'Data Sources' tree with 'cfreds_2015_data_leakage_rm#2.dd,1 Host' selected. Under 'File Views', 'Deleted Files' is expanded, showing 'File System (0)', 'All (62)', and 'MB File Size' (MB 50 - 200MB (0), MB 200MB - 1GB (0), MB 1GB+ (0)). The 'Data Artifacts' section is expanded, showing 'Metadata (25)', 'Analysis Results', 'EXIF Metadata (11)', 'Extension Mismatch Detected (2)', 'Keyword Hits (28)', 'User Content Suspected (11)', 'OS Accounts', 'Tags', 'Score', and 'Reports'. The main pane shows a table of file carving results with columns: Source Name, S, C, O, Description, User ID, Program Name, and Owner. The table lists 25 results, including files like 'secret_project_technical_review_3.ppt', 'secret_project_technical_review_3.doc', 'secret_project_technical_review_#1', 'secret_project_technical_review_#2', 'Microsoft PowerPoint_97-2003_Presentation1.ppt', 'oleObject7.bin', 'oleObject2.bin', 'oleObject5.bin', 'oleObject6.bin', 'oleObject3.bin', 'oleObject1.bin', 'secret_project_technical_review_#1', 'secret_project_proposal', and 'oleObject11.bin'. The bottom status bar shows the time as 10:31 PM on 2025-11-03.

Source Name	S	C	O	Description	User ID	Program Name	Owner
cfreds_2015_data_leakage_rm#2.dd,1 Host							
cfreds_2015_data_leakage_rm#2.dd							
\$CarvedFiles (1)							
File Types							
Deleted Files							
File System (0)							
All (62)							
MB File Size							
MB 50 - 200MB (0)							
MB 200MB - 1GB (0)							
MB 1GB+ (0)							
Data Artifacts							
Metadata (25)							
Analysis Results							
EXIF Metadata (11)							
Extension Mismatch Detected (2)							
Keyword Hits (28)							
User Content Suspected (11)							
OS Accounts							
Tags							
Score							
Reports							

2. List all files that were opened in 'RM#2'.

Ans:

Timeline - Editor

Timeline x

Display Times In: ☒ Local Time Zone ☐ GMT / UTC

History Back Forward

Filters

Apply Reset

Must include text: enter filter string

☐ Must be tagged

☐ Must have hash hit

☐ Limit data sources to

☐ cfreds2015_data_leakage...

☐ Limit file types to

☐ Media

☐ Documents

☐ Executables

☐ Other

☐ Limit event types to

☐ File System

☐ File Accessed

☐ File Changed

☐ File Created

☐ File Modified

☐ File Activity

☐ Web Activity

☐ Web Bookmarks

View Mode: Counts Details List

Add Event Snapshot Report Refresh View

55 events

Date/Time	Event Type	Description	Tagged	Hash Hit
2001-12-12 16:10:54	Document Created	Document Created :		
2003-02-04 20:11:17	Document Created	Document Created :		
2003-07-10 13:33:15	Document Created	Document Created :		
2003-07-10 15:17:56	Document Created	Document Created :		
2003-07-10 17:37:13	Document Last Saved	Document Last Saved :		
2003-09-24 15:33:42	Exif	Eastman Kodak Company : KODAK DIGITAL SCIENCE DC260 (V01.00) : image_13.jpg		
2003-11-25 00:09:57	Document Created	Document Created :		
2003-12-10 17:27:44	Exif	Eastman Kodak Company : KODAK DIGITAL SCIENCE DC260 (V01.00) : image_113.jpg		
2004-02-20 06:53:23	Document Created	Document Created :		
2004-05-06 01:10:13	Document Created	Document Created :		
2004-10-05 14:50:21	Document Created	Document Created :		
2004-10-06 10:46:47	Exif	EASTMAN KODAK COMPANY : KODAK DX4530 ZOOM DIGITAL CAMERA : f0439240.jpg		
2004-10-07 07:24:19	Exif	EASTMAN KODAK COMPANY : KODAK DX4530 ZOOM DIGITAL CAMERA : f0508296.jpg		
2004-10-07 11:55:46	Exif	EASTMAN KODAK COMPANY : KODAK DX4530 ZOOM DIGITAL CAMERA : f0506536.jpg		
2005-04-04 04:00:33	Exif	EASTMAN KODAK COMPANY : KODAK DX4530 ZOOM DIGITAL CAMERA : f0504872.jpg		
2005-04-04 05:41:46	Exif	EASTMAN KODAK COMPANY : KODAK DX4530 ZOOM DIGITAL CAMERA : f0502376.jpg		
2005-12-03 16:53:25	Document Created	Document Created :		
2006-01-05 16:52:42	Document Last Saved	Document Last Saved :		
2006-03-13 17:15:18	Document Last Saved	Document Last Saved :		
2006-03-31 10:00:00	Document Created	Document Created :		

Start: Dec 12, 2001, 4:10:54 PM Jump By: year End: Feb 6, 2015, 8:04:01 PM

Hex Text Application File Metadata OS Account Data Artifacts Analysis Results Context Annotations Other Occurrences

10:41 PM 2025-11-03

3. Recover deleted files from USB drive 'RM#2'. What files were you able to recover?

Ans: Most images could be opened, but MP4 files could not.

The screenshot displays the Autopsy 4.22.1 interface, showing the results of a file recovery operation on a USB drive labeled 'RM#2'. The interface is divided into several panes:

- Left Pane:** Contains a tree view of the file system, including 'Data Sources', 'File Views', 'File Types', 'Deleted Files', 'MB File Size', 'Data Artifacts', 'Metadata (25)', 'Analysis Results', 'EXIF Metadata (11)', 'Extension Mismatch Detected (2)', 'Keyword Hits (28)', 'User Content Suspected (11)', 'OS Accounts', 'Tags', 'Score', and 'Reports'.
- Top Pane:** Shows the 'Listing' tab with a table of recovered files. The table has columns for Name, S, C, O, Modified Time, Change Time, Access Time, Created Time, Size, Flags(Dir), and File. The files listed include various image formats (gif, png, jpg, bmp, tif) and video formats (wmv, mov, mp4, fat). A tooltip indicates that there were 1 data source(s) found with occurrences of the MD5 correlation value 0-00 00:00:00.
- Bottom Pane:** Displays the 'File Metadata' tab for the selected file 'f0462760.gif'. It shows a thumbnail of the image, which appears to be a collection of colorful, cylindrical objects. Below the thumbnail, there is a 'Tags Menu' and a 'Playback' section for deleted videos, which is currently empty and displays a message: 'Playback of deleted videos is not supported, use an external player.'

The interface also shows a Windows taskbar at the bottom with the system clock set to 10:42 PM on 2025-11-03.

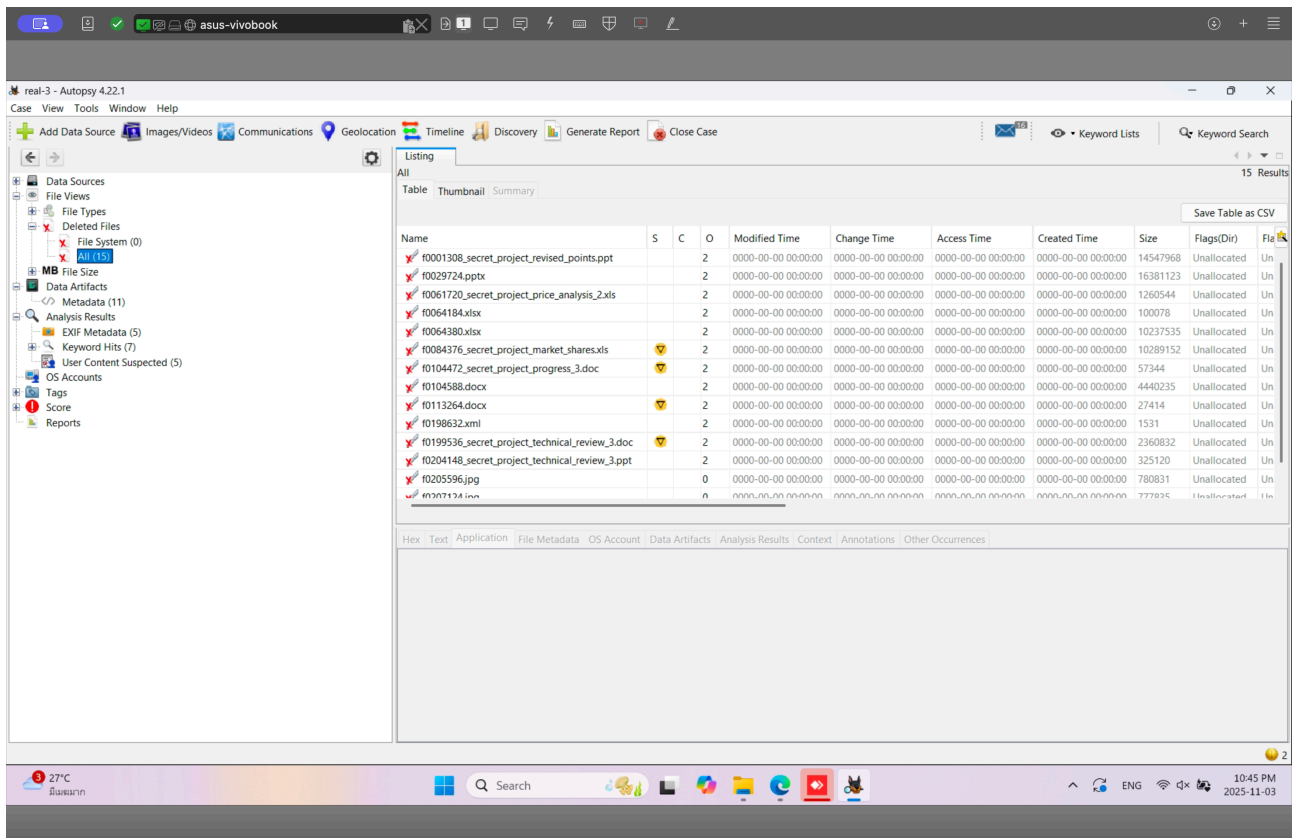
4. What actions were performed for anti-forensics on USB drive 'RM#2'? [Hint: this can be inferred from the results of the above question]

Ans:

There was an attempt to delete the metadata, and some file formats were tampered with. Notably, there were files with strange names ending in .txt, but the content inside didn't seem to be text, so I tried changing the type to .ppt.

5. Recover hidden files from the CD-R 'RM#3'. What files were you able to recover?

Ans: Found 15 files.



6. What actions were performed for anti-forensics (data hiding) on CD-R 'RM#3'?

Ans: It seems that only the metadata was deleted or destroyed.